MSG No. 129.421 P_C4/nmm [Type IV, tetragonal]

Table 1: Wyckoff site: 2a, site symmetry: 4/m'mm

No.	position	mapping
1	$\left[\frac{3}{4}, \frac{3}{4}, 0\right]$	[1,2,3,6,12,13,15,16,20,21,23,24,25,26,27,30]
2	$\left[\frac{1}{4}, \frac{1}{4}, 0\right]$	[4,5,7,8,9,10,11,14,17,18,19,22,28,29,31,32]

Table 2: Wyckoff site: 2b, site symmetry: 4/m'mm

No.	position	mapping
1	$\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{2}\right]$	[1,2,3,6,12,13,15,16,20,21,23,24,25,26,27,30]
2	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{2}\right]$	[4,5,7,8,9,10,11,14,17,18,19,22,28,29,31,32]

Table 3: Wyckoff site: 2c, site symmetry: 4'/m'mm'

No.	position	mapping
1	$\left[\frac{3}{4}, \frac{1}{4}, 0\right]$	[1,6,7,8,10,11,12,13,18,19,20,21,25,30,31,32]
2	$\left[\frac{1}{4}, \frac{3}{4}, 0\right]$	[2,3,4,5,9,14,15,16,17,22,23,24,26,27,28,29]

Table 4: Wyckoff site: 2d, site symmetry: 4'/m'mm'

No.	position	mapping
1	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{2}\right]$	[1,6,7,8,10,11,12,13,18,19,20,21,25,30,31,32]
2	$\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{2}\right]$	[2,3,4,5,9,14,15,16,17,22,23,24,26,27,28,29]

Table 5: Wyckoff site: 4e, site symmetry: m'.mm'

No.	position	mapping
1	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	[1,7,9,15,22,24,30,32]
2	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	[2,5,10,13,19,20,27,28]
3	$\left[0, 0, \frac{1}{2}\right]$	[3,4,11,12,18,21,26,29]
4	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	[6,8,14,16,17,23,25,31]

Table 6: Wyckoff site: 4f, site symmetry: m'.mm'

No.	position	mapping
1	$\left[\frac{1}{2}, 0, 0\right]$	[1,7,9,15,22,24,30,32]
2	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	[2,5,10,13,19,20,27,28]
3	$[0, 0, 0]$	[3,4,11,12,18,21,26,29]
4	$\left[0, \frac{1}{2}, 0\right]$	[6,8,14,16,17,23,25,31]

Table 7: Wyckoff site: `4g`, site symmetry: `4mm`

No.	position	mapping
1	$\left[\frac{3}{4}, \frac{3}{4}, z\right]$	`[1,2,3,6,12,13,15,16]`
2	$\left[\frac{1}{4}, \frac{1}{4}, -z\right]$	`[4,5,7,8,9,10,11,14]`
3	$\left[\frac{1}{4}, \frac{1}{4}, z\right]$	`[17,18,19,22,28,29,31,32]`
4	$\left[\frac{3}{4}, \frac{3}{4}, -z\right]$	`[20,21,23,24,25,26,27,30]`

Table 8: Wyckoff site: `4h`, site symmetry: `4'mm'`

No.	position	mapping
1	$\left[\frac{3}{4}, \frac{1}{4}, z\right]$	`[1,6,12,13,18,19,31,32]`
2	$\left[\frac{1}{4}, \frac{3}{4}, z\right]$	`[2,3,15,16,17,22,28,29]`
3	$\left[\frac{1}{4}, \frac{3}{4}, -z\right]$	`[4,5,9,14,23,24,26,27]`
4	$\left[\frac{3}{4}, \frac{1}{4}, -z\right]$	`[7,8,10,11,20,21,25,30]`

Table 9: Wyckoff site: `8i`, site symmetry: `2.mm`

No.	position	mapping
1	$\left[\frac{1}{2}, 0, z\right]$	`[1,15,22,32]`
2	$\left[\frac{1}{2}, \frac{1}{2}, z\right]$	`[2,13,19,28]`
3	$[0, 0, z]$	`[3,12,18,29]`
4	$[0, 0, -z]$	`[4,11,21,26]`
5	$\left[\frac{1}{2}, \frac{1}{2}, -z\right]$	`[5,10,20,27]`
6	$\left[0, \frac{1}{2}, z\right]$	`[6,16,17,31]`
7	$\left[\frac{1}{2}, 0, -z\right]$	`[7,9,24,30]`
8	$\left[0, \frac{1}{2}, -z\right]$	`[8,14,23,25]`

Table 10: Wyckoff site: `8j`, site symmetry: `m'm2'`.

No.	position	mapping
1	$\left[\frac{3}{4}, y, 0\right]$	`[1,12,21,30]`
2	$\left[\frac{1}{2} - y, \frac{3}{4}, 0\right]$	`[2,15,24,27]`
3	$\left[y, \frac{3}{4}, 0\right]$	`[3,16,23,26]`
4	$\left[\frac{1}{4}, -y, 0\right]$	`[4,9,22,29]`
5	$\left[\frac{1}{4}, y + \frac{1}{2}, 0\right]$	`[5,14,17,28]`
6	$\left[\frac{3}{4}, \frac{1}{2} - y, 0\right]$	`[6,13,20,25]`
7	$\left[y + \frac{1}{2}, \frac{1}{4}, 0\right]$	`[7,10,19,32]`
8	$\left[-y, \frac{1}{4}, 0\right]$	`[8,11,18,31]`

Table 11: Wyckoff site: `8k`, site symmetry: `m'm2'`.

No.	position	mapping
1	$\left[\frac{3}{4}, y, \frac{1}{2}\right]$	`[1,12,21,30]`
2	$\left[\frac{1}{2} - y, \frac{3}{4}, \frac{1}{2}\right]$	`[2,15,24,27]`

continued ...

Table 11

No.	position	mapping
3	$\left[y,\ \frac{3}{4},\ \frac{1}{2}\right]$	`[3,16,23,26]`
4	$\left[\frac{1}{4},\ -y,\ \frac{1}{2}\right]$	`[4,9,22,29]`
5	$\left[\frac{1}{4},\ y+\frac{1}{2},\ \frac{1}{2}\right]$	`[5,14,17,28]`
6	$\left[\frac{3}{4},\ \frac{1}{2}-y,\ \frac{1}{2}\right]$	`[6,13,20,25]`
7	$\left[y+\frac{1}{2},\ \frac{1}{4},\ \frac{1}{2}\right]$	`[7,10,19,32]`
8	$\left[-y,\ \frac{1}{4},\ \frac{1}{2}\right]$	`[8,11,18,31]`

Table 12: Wyckoff site: `8l`, site symmetry: `m'.2'm`

No.	position	mapping
1	$[x,\ x,\ 0]$	`[1,16,23,30]`
2	$\left[\frac{1}{2}-x,\ x,\ 0\right]$	`[2,12,21,27]`
3	$\left[x,\ \frac{1}{2}-x,\ 0\right]$	`[3,13,20,26]`
4	$\left[x+\frac{1}{2},\ -x,\ 0\right]$	`[4,10,19,29]`
5	$\left[-x,\ x+\frac{1}{2},\ 0\right]$	`[5,11,18,28]`
6	$\left[\frac{1}{2}-x,\ \frac{1}{2}-x,\ 0\right]$	`[6,15,24,25]`
7	$\left[x+\frac{1}{2},\ x+\frac{1}{2},\ 0\right]$	`[7,14,17,32]`
8	$[-x,\ -x,\ 0]$	`[8,9,22,31]`

Table 13: Wyckoff site: `8m`, site symmetry: `m'.2'm`

No.	position	mapping
1	$\left[x,\ x,\ \frac{1}{2}\right]$	`[1,16,23,30]`
2	$\left[\frac{1}{2}-x,\ x,\ \frac{1}{2}\right]$	`[2,12,21,27]`
3	$\left[x,\ \frac{1}{2}-x,\ \frac{1}{2}\right]$	`[3,13,20,26]`
4	$\left[x+\frac{1}{2},\ -x,\ \frac{1}{2}\right]$	`[4,10,19,29]`
5	$\left[-x,\ x+\frac{1}{2},\ \frac{1}{2}\right]$	`[5,11,18,28]`
6	$\left[\frac{1}{2}-x,\ \frac{1}{2}-x,\ \frac{1}{2}\right]$	`[6,15,24,25]`
7	$\left[x+\frac{1}{2},\ x+\frac{1}{2},\ \frac{1}{2}\right]$	`[7,14,17,32]`
8	$\left[-x,\ -x,\ \frac{1}{2}\right]$	`[8,9,22,31]`

Table 14: Wyckoff site: `8n`, site symmetry: `m'.2m'`

No.	position	mapping
1	$\left[x,\ x+\frac{1}{2},\ 0\right]$	`[1,7,30,32]`
2	$[-x,\ x,\ 0]$	`[2,5,27,28]`
3	$\left[x+\frac{1}{2},\ \frac{1}{2}-x,\ 0\right]$	`[3,4,26,29]`
4	$\left[\frac{1}{2}-x,\ -x,\ 0\right]$	`[6,8,25,31]`
5	$\left[-x,\ \frac{1}{2}-x,\ 0\right]$	`[9,15,22,24]`
6	$[x,\ -x,\ 0]$	`[10,13,19,20]`
7	$\left[\frac{1}{2}-x,\ x+\frac{1}{2},\ 0\right]$	`[11,12,18,21]`
8	$\left[x+\frac{1}{2},\ x,\ 0\right]$	`[14,16,17,23]`

Table 15: Wyckoff site: 8o, site symmetry: m'.2m'

No.	position	mapping
1	$\left[x,\, x+\frac{1}{2},\, \frac{1}{2}\right]$	[1,7,30,32]
2	$\left[-x,\, x,\, \frac{1}{2}\right]$	[2,5,27,28]
3	$\left[x+\frac{1}{2},\, \frac{1}{2}-x,\, \frac{1}{2}\right]$	[3,4,26,29]
4	$\left[\frac{1}{2}-x,\, -x,\, \frac{1}{2}\right]$	[6,8,25,31]
5	$\left[-x,\, \frac{1}{2}-x,\, \frac{1}{2}\right]$	[9,15,22,24]
6	$\left[x,\, -x,\, \frac{1}{2}\right]$	[10,13,19,20]
7	$\left[\frac{1}{2}-x,\, x+\frac{1}{2},\, \frac{1}{2}\right]$	[11,12,18,21]
8	$\left[x+\frac{1}{2},\, x,\, \frac{1}{2}\right]$	[14,16,17,23]

Table 16: Wyckoff site: 16p, site symmetry: m'..

No.	position	mapping
1	$[x,\, y,\, 0]$	[1,30]
2	$\left[\frac{1}{2}-y,\, x,\, 0\right]$	[2,27]
3	$\left[y,\, \frac{1}{2}-x,\, 0\right]$	[3,26]
4	$\left[x+\frac{1}{2},\, -y,\, 0\right]$	[4,29]
5	$\left[-x,\, y+\frac{1}{2},\, 0\right]$	[5,28]
6	$\left[\frac{1}{2}-x,\, \frac{1}{2}-y,\, 0\right]$	[6,25]
7	$\left[y+\frac{1}{2},\, x+\frac{1}{2},\, 0\right]$	[7,32]
8	$[-y,\, -x,\, 0]$	[8,31]
9	$[-x,\, -y,\, 0]$	[9,22]
10	$\left[y+\frac{1}{2},\, -x,\, 0\right]$	[10,19]
11	$\left[-y,\, x+\frac{1}{2},\, 0\right]$	[11,18]
12	$\left[\frac{1}{2}-x,\, y,\, 0\right]$	[12,21]
13	$\left[x,\, \frac{1}{2}-y,\, 0\right]$	[13,20]
14	$\left[x+\frac{1}{2},\, y+\frac{1}{2},\, 0\right]$	[14,17]
15	$\left[\frac{1}{2}-y,\, \frac{1}{2}-x,\, 0\right]$	[15,24]
16	$[y,\, x,\, 0]$	[16,23]

Table 17: Wyckoff site: 16q, site symmetry: m'..

No.	position	mapping
1	$\left[x,\, y,\, \frac{1}{2}\right]$	[1,30]
2	$\left[\frac{1}{2}-y,\, x,\, \frac{1}{2}\right]$	[2,27]
3	$\left[y,\, \frac{1}{2}-x,\, \frac{1}{2}\right]$	[3,26]
4	$\left[x+\frac{1}{2},\, -y,\, \frac{1}{2}\right]$	[4,29]
5	$\left[-x,\, y+\frac{1}{2},\, \frac{1}{2}\right]$	[5,28]
6	$\left[\frac{1}{2}-x,\, \frac{1}{2}-y,\, \frac{1}{2}\right]$	[6,25]
7	$\left[y+\frac{1}{2},\, x+\frac{1}{2},\, \frac{1}{2}\right]$	[7,32]
8	$\left[-y,\, -x,\, \frac{1}{2}\right]$	[8,31]
9	$\left[-x,\, -y,\, \frac{1}{2}\right]$	[9,22]
10	$\left[y+\frac{1}{2},\, -x,\, \frac{1}{2}\right]$	[10,19]
11	$\left[-y,\, x+\frac{1}{2},\, \frac{1}{2}\right]$	[11,18]
12	$\left[\frac{1}{2}-x,\, y,\, \frac{1}{2}\right]$	[12,21]

continued ...

Table 17

No.	position	mapping
13	$\left[x,\ \frac{1}{2}-y,\ \frac{1}{2}\right]$	[13,20]
14	$\left[x+\frac{1}{2},\ y+\frac{1}{2},\ \frac{1}{2}\right]$	[14,17]
15	$\left[\frac{1}{2}-y,\ \frac{1}{2}-x,\ \frac{1}{2}\right]$	[15,24]
16	$\left[y,\ x,\ \frac{1}{2}\right]$	[16,23]

Table 18: Wyckoff site: 16r, site symmetry: .m.

No.	position	mapping
1	$\left[\frac{3}{4},\ y,\ z\right]$	[1,12]
2	$\left[\frac{1}{2}-y,\ \frac{3}{4},\ z\right]$	[2,15]
3	$\left[y,\ \frac{3}{4},\ z\right]$	[3,16]
4	$\left[\frac{1}{4},\ -y,\ -z\right]$	[4,9]
5	$\left[\frac{1}{4},\ y+\frac{1}{2},\ -z\right]$	[5,14]
6	$\left[\frac{3}{4},\ \frac{1}{2}-y,\ z\right]$	[6,13]
7	$\left[y+\frac{1}{2},\ \frac{1}{4},\ -z\right]$	[7,10]
8	$\left[-y,\ \frac{1}{4},\ -z\right]$	[8,11]
9	$\left[\frac{1}{4},\ y+\frac{1}{2},\ z\right]$	[17,28]
10	$\left[-y,\ \frac{1}{4},\ z\right]$	[18,31]
11	$\left[y+\frac{1}{2},\ \frac{1}{4},\ z\right]$	[19,32]
12	$\left[\frac{3}{4},\ \frac{1}{2}-y,\ -z\right]$	[20,25]
13	$\left[\frac{3}{4},\ y,\ -z\right]$	[21,30]
14	$\left[\frac{1}{4},\ -y,\ z\right]$	[22,29]
15	$\left[y,\ \frac{3}{4},\ -z\right]$	[23,26]
16	$\left[\frac{1}{2}-y,\ \frac{3}{4},\ -z\right]$	[24,27]

Table 19: Wyckoff site: 16s, site symmetry: ..m

No.	position	mapping
1	$[x,\ x,\ z]$	[1,16]
2	$\left[\frac{1}{2}-x,\ x,\ z\right]$	[2,12]
3	$\left[x,\ \frac{1}{2}-x,\ z\right]$	[3,13]
4	$\left[x+\frac{1}{2},\ -x,\ -z\right]$	[4,10]
5	$\left[-x,\ x+\frac{1}{2},\ -z\right]$	[5,11]
6	$\left[\frac{1}{2}-x,\ \frac{1}{2}-x,\ z\right]$	[6,15]
7	$\left[x+\frac{1}{2},\ x+\frac{1}{2},\ -z\right]$	[7,14]
8	$[-x,\ -x,\ -z]$	[8,9]
9	$\left[x+\frac{1}{2},\ x+\frac{1}{2},\ z\right]$	[17,32]
10	$\left[-x,\ x+\frac{1}{2},\ z\right]$	[18,28]
11	$\left[x+\frac{1}{2},\ -x,\ z\right]$	[19,29]
12	$\left[x,\ \frac{1}{2}-x,\ -z\right]$	[20,26]
13	$\left[\frac{1}{2}-x,\ x,\ -z\right]$	[21,27]
14	$[-x,\ -x,\ z]$	[22,31]
15	$[x,\ x,\ -z]$	[23,30]
16	$\left[\frac{1}{2}-x,\ \frac{1}{2}-x,\ -z\right]$	[24,25]

Table 20: Wyckoff site: `16t`, site symmetry: `..m'`

No.	position	mapping
1	$\left[x,\ x+\frac{1}{2},\ z\right]$	`[1,32]`
2	$\left[-x,\ x,\ z\right]$	`[2,28]`
3	$\left[x+\frac{1}{2},\ \frac{1}{2}-x,\ z\right]$	`[3,29]`
4	$\left[x+\frac{1}{2},\ \frac{1}{2}-x,\ -z\right]$	`[4,26]`
5	$\left[-x,\ x,\ -z\right]$	`[5,27]`
6	$\left[\frac{1}{2}-x,\ -x,\ z\right]$	`[6,31]`
7	$\left[x,\ x+\frac{1}{2},\ -z\right]$	`[7,30]`
8	$\left[\frac{1}{2}-x,\ -x,\ -z\right]$	`[8,25]`
9	$\left[-x,\ \frac{1}{2}-x,\ -z\right]$	`[9,24]`
10	$\left[x,\ -x,\ -z\right]$	`[10,20]`
11	$\left[\frac{1}{2}-x,\ x+\frac{1}{2},\ -z\right]$	`[11,21]`
12	$\left[\frac{1}{2}-x,\ x+\frac{1}{2},\ z\right]$	`[12,18]`
13	$\left[x,\ -x,\ z\right]$	`[13,19]`
14	$\left[x+\frac{1}{2},\ x,\ -z\right]$	`[14,23]`
15	$\left[-x,\ \frac{1}{2}-x,\ z\right]$	`[15,22]`
16	$\left[x+\frac{1}{2},\ x,\ z\right]$	`[16,17]`

Table 21: Wyckoff site: `32u`, site symmetry: `1`

No.	position	mapping
1	$\left[x,\ y,\ z\right]$	`[1]`
2	$\left[\frac{1}{2}-y,\ x,\ z\right]$	`[2]`
3	$\left[y,\ \frac{1}{2}-x,\ z\right]$	`[3]`
4	$\left[x+\frac{1}{2},\ -y,\ -z\right]$	`[4]`
5	$\left[-x,\ y+\frac{1}{2},\ -z\right]$	`[5]`
6	$\left[\frac{1}{2}-x,\ \frac{1}{2}-y,\ z\right]$	`[6]`
7	$\left[y+\frac{1}{2},\ x+\frac{1}{2},\ -z\right]$	`[7]`
8	$\left[-y,\ -x,\ -z\right]$	`[8]`
9	$\left[-x,\ -y,\ -z\right]$	`[9]`
10	$\left[y+\frac{1}{2},\ -x,\ -z\right]$	`[10]`
11	$\left[-y,\ x+\frac{1}{2},\ -z\right]$	`[11]`
12	$\left[\frac{1}{2}-x,\ y,\ z\right]$	`[12]`
13	$\left[x,\ \frac{1}{2}-y,\ z\right]$	`[13]`
14	$\left[x+\frac{1}{2},\ y+\frac{1}{2},\ -z\right]$	`[14]`
15	$\left[\frac{1}{2}-y,\ \frac{1}{2}-x,\ z\right]$	`[15]`
16	$\left[y,\ x,\ z\right]$	`[16]`
17	$\left[x+\frac{1}{2},\ y+\frac{1}{2},\ z\right]$	`[17]`
18	$\left[-y,\ x+\frac{1}{2},\ z\right]$	`[18]`
19	$\left[y+\frac{1}{2},\ -x,\ z\right]$	`[19]`
20	$\left[x,\ \frac{1}{2}-y,\ -z\right]$	`[20]`
21	$\left[\frac{1}{2}-x,\ y,\ -z\right]$	`[21]`
22	$\left[-x,\ -y,\ z\right]$	`[22]`
23	$\left[y,\ x,\ -z\right]$	`[23]`
24	$\left[\frac{1}{2}-y,\ \frac{1}{2}-x,\ -z\right]$	`[24]`
25	$\left[\frac{1}{2}-x,\ \frac{1}{2}-y,\ -z\right]$	`[25]`

continued ...

Table 21

No.	position	mapping
26	$\left[y,\, \frac{1}{2}-x,\, -z\right]$	[26]
27	$\left[\frac{1}{2}-y,\, x,\, -z\right]$	[27]
28	$\left[-x,\, y+\frac{1}{2},\, z\right]$	[28]
29	$\left[x+\frac{1}{2},\, -y,\, z\right]$	[29]
30	$[x,\, y,\, -z]$	[30]
31	$[-y,\, -x,\, z]$	[31]
32	$\left[y+\frac{1}{2},\, x+\frac{1}{2},\, z\right]$	[32]